Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Right-triangle trigonometry**

In a right triangle, the angle θ has an opposite side of 7 and hypotenuse 25. Find $\cos\theta$ and $\tan\theta$.

Adjacent side: $\sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$. $\cos\theta = \frac{24}{25}$. $\tan\theta = \frac{7}{24}$.

2 **[Foundational]** • **Target: Right-triangle trigonometry**

In a right triangle, angle θ has an adjacent side of 8 and opposite side 15. Find $\sin\theta$ and the hypotenuse.

Hypotenuse: $\sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17$. $\sin\theta = \frac{15}{17}$.

3 **[Foundational]** • **Target: Exact values and the unit circle**

State the exact value of $\sin\left(\frac{\pi}{3}\right)$ and $\cos(45^\circ)$.

$\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$. $\cos(45^\circ) = \frac{\sqrt{2}}{2}$.

4 **[Foundational]** • **Target: Graphs of trigonometric functions**

State the amplitude and period of $y = 4\sin(2x)$.

Amplitude: 4. Period: $\frac{2\pi}{2} = \pi$.

5 **[Foundational]** • **Target: Trigonometric identities**

Given $\sin\theta = \frac{3}{5}$ with θ acute, find $\cos\theta$ using the identity $\sin^2\theta + \cos^2\theta = 1$.

$\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$. Since θ is acute, $\cos\theta = \frac{4}{5}$.

6 **[Foundational]** • **Target: Law of Sines and Law of Cosines**

In triangle ABC, $a = 9$, $b = 12$, and angle $C = 55^\circ$. Find side c, to the nearest tenth, using the Law of Cosines.

$c^2 = 9^2 + 12^2 - 2(9)(12)\cos(55^\circ) = 81 + 144 - 216(0.5736) \approx 225 - 123.9 \approx 101.1$. $c \approx \sqrt{101.1} \approx 10.1$.

7 **[Foundational]** • **Target: Law of Sines and Law of Cosines**

In triangle ABC, angle $A = 40^\circ$, angle $B = 70^\circ$, and $a = 8$. Find side b, to the nearest tenth, using the Law of Sines.

$\frac{a}{\sin A} = \frac{b}{\sin B}$: $\frac{8}{\sin 40^\circ} = \frac{b}{\sin 70^\circ}$. $b = \frac{8\sin 70^\circ}{\sin 40^\circ} \approx \frac{8(0.9397)}{0.6428} \approx 11.7$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Law of Sines and Law of Cosines**

In triangle ABC, $a = 7$, $b = 10$, $c = 13$. Find the largest angle, to the nearest degree, using the Law of Cosines.

The largest angle is opposite the longest side, so we find angle C:
$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{49 + 100 - 169}{2(7)(10)} = \frac{-20}{140} \approx -0.1429$. $C = \cos^{-1}(-0.1429) \approx 98^\circ$.

9 **[Intermediate]** • **Target: Trigonometric identities for compound angles**

Use the identity $\sin(A + B) = \sin A \cos B + \cos A \sin B$ to find the exact value of $\sin(75^\circ)$, using $75^\circ = 45^\circ + 30^\circ$.

$\sin(75^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}$.

10 **[Intermediate]** • **Target: A well-designed word problem: the height of a lighthouse**

This question has two parts. A ship observes a lighthouse at an angle of elevation of 18°. After sailing 500 meters directly toward the lighthouse, the angle of elevation increases to 30°. (a) Let h be the lighthouse's height and d the final distance from its base. Write two equations relating h and d using the tangent ratio. (b) Solve the system to find h, to the nearest meter.

(a) At the closer position: $\tan(30^\circ) = \frac{h}{d}$, so $h = d\tan(30^\circ)$. At the farther position (distance $d + 500$): $\tan(18^\circ) = \frac{h}{d + 500}$, so $h = (d + 500)\tan(18^\circ)$. (b) Setting equal: $d\tan(30^\circ) = (d + 500)\tan(18^\circ)$. Using $\tan(30^\circ) \approx 0.5774$ and $\tan(18^\circ) \approx 0.3249$: $0.5774d = 0.3249d + 162.45$, so $0.2525d = 162.45$, giving $d \approx 643.4$. Then $h = 643.4 \times 0.5774 \approx 372$ meters.

11 **[Intermediate]** • **Target: Graphs of trigonometric functions**

[Functions: Modulus & Inverse · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

12 **[Intermediate]** • **Target: Trigonometric identities**

[Advanced Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

13 **[Intermediate]** • **Target: Solving trigonometric equations**

[Functions, Modulus & Quadratic Theory · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

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14 **[Intermediate]** • **Target: Law of Sines and Law of Cosines**

[Trigonometry & Circular Measure · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Trigonometric identities for compound angles** • **Cross-Lesson: 19.01 Normal Distribution**

[Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

16 **[Challenging]** • **Target: A well-designed word problem: the height of a lighthouse** • **Cross-Lesson: 2.01 Simultaneous Equations & Inequalities**

[Rational Functions, Limits & Continuity · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

17 **[Challenging]** • **Target: Right-triangle trigonometry** • **Cross-Lesson: 3.01 Graphs & Transformations**

Given $f(x) = 2x - 3$ for $x \in \mathbb{R}$, sketch $y = |f(x)|$ and solve $|2x - 3| = 5$.

Sketch: V-shape with vertex at $(1.5, 0)$. Solve: $2x - 3 = 5 \Rightarrow x = 4$ or $2x - 3 = -5 \Rightarrow x = -1$.

18 **[Challenging]** • **Target: Exact values and the unit circle** • **Cross-Lesson: 4.01 Coordinate Geometry**

Express $3\sin\theta + 4\cos\theta$ in the form $R\sin(\theta + \alpha)$ where $R > 0$ and $0 < \alpha < 90°$.

$R = \sqrt{9 + 16} = 5$. $\tan\alpha = 4/3 \Rightarrow \alpha = 53.13°$. So $5\sin(\theta + 53.13°)$.

19 **[Challenging]** • **Target: Graphs of trigonometric functions** • **Cross-Lesson: 6.01 Differentiation & Integration**

Show that $x^3 + 2x - 6 = 0$ has a root in $[1, 2]$. Taking $x_0 = 1.5$, perform one Newton-Raphson iteration.

$f(1) = -3 < 0$, $f(2) = 6 > 0$: sign change confirms root. $f(1.5) = 1.5^3 + 3 - 6 = -0.625$.
$f'(x) = 3x^2 + 2 \Rightarrow f'(1.5) = 8.75$. $x_1 = 1.5 - \frac{-0.625}{8.75} \approx 1.571$.

20 **[Challenging]** • **Target: Trigonometric identities** • **Cross-Lesson: 7.01 Statistics & Probability**

Express $\frac{3x + 1}{(x + 1)(x - 2)}$ in partial fractions.

$\frac{A}{x+1} + \frac{B}{x-2}$. Cover-up: let $x = 2$: $B = \frac{7}{3}$. Let $x = -1$: $A = \frac{-2}{-3} = \frac{2}{3}$. Answer: $\frac{2/3}{x+1} + \frac{7/3}{x-2}$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Right-triangle trigonometry

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?

a. $\cos\theta$ ✓

b. $\sin\theta$

c. $\tan\theta$

d. 1

By definition, the coordinates of a point $P(x, y)$ on the unit circle corresponding to an angle θ (measured counterclockwise from the positive x-axis) are given by $(\cos\theta, \sin\theta)$. Therefore, the x-coordinate of the point P is $\cos\theta$.

22 [Foundational] • Target: Exact values and the unit circle

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

a. $\frac{7}{25}$

b. $-\frac{7}{25}$ ✓

c. $\frac{24}{25}$

d. $-\frac{24}{25}$

We are given $\cos\alpha = \frac{3}{5}$ and α is in the first quadrant. Using the identity $\sin^2\alpha + \cos^2\alpha = 1$, we find $\sin\alpha = \sqrt{1 - (\frac{3}{5})^2} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$ (since α is in the first quadrant, $\sin\alpha$ is positive). We are given $\sin\beta = -\frac{4}{5}$ and β is in the third quadrant. Using the identity $\sin^2\beta + \cos^2\beta = 1$, we find.

23 [Foundational] • Target: Graphs of trigonometric functions

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$ ✓

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

The problem states that point $P(x, y)$ lies on the unit circle, meaning $x^2 + y^2 = 1$. We are given that the y-coordinate of P is $y = \frac{3}{5}$. Substituting this into the unit circle equation, we get $x^2 + (\frac{3}{5})^2 = 1$. This simplifies to $x^2 + \frac{9}{25} = 1$. Subtracting $\frac{9}{25}$ from both sides gives $x^2 = 1 - \frac{9}{25} = \frac{16}{25}$. Taking the square root of both sides, we get $x = \pm\frac{4}{5}$. The problem also states that point P is in the second quadrant, where the x.

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24 [Foundational] • Target: Trigonometric identities

Consider a point P on the unit circle in the second quadrant such that the y-coordinate of P is $\frac{3}{5}$. Let θ be the angle in standard position whose terminal side passes through P. Find the value of $\cos\theta$.

a. $-\frac{3}{5}$

b. $\frac{3}{5}$

c. $-\frac{4}{5}$ ✓

d. $\frac{4}{5}$

On the unit circle, any point P(x, y) satisfies the equation $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $\frac{3}{5}$ and P is in the second quadrant. For a point $P(\cos\theta, \sin\theta)$ on the unit circle, we have $x = \cos\theta$ and $y = \sin\theta$. Thus, we have $\sin\theta = \frac{3}{5}$. Substituting this into the unit circle equation: $\cos^2\theta + (\frac{3}{5})^2 = 1$. $\cos^2\theta + \frac{9}{25} = 1$. $\cos^2\theta = 1 - \frac{9}{25}$. $\cos^2\theta = \frac{16}{25}$..

25 [Foundational] • Target: Solving trigonometric equations

Consider a point P on the unit circle in standard position. If the angle θ in standard position has a terminal side passing through P such that $\cos(\theta) = \frac{1}{2}$ and $\sin(\theta) > 0$, find the value of $\sin(\theta)$.

a. $\frac{\sqrt{3}}{2}$ ✓

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. -1

For any point on the unit circle, the coordinates are $(\cos(\theta), \sin(\theta))$. The equation of the unit circle is $x^2 + y^2 = 1$. Substituting the coordinates, we get $\cos^2(\theta) + \sin^2(\theta) = 1$. We are given that $\cos(\theta) = \frac{1}{2}$. So, $\left(\frac{1}{2}\right)^2 + \sin^2(\theta) = 1$. This simplifies to $\frac{1}{4} + \sin^2(\theta) = 1$. Subtracting $\frac{1}{4}$ from both sides gives $\sin^2(\theta) = 1 - \frac{1}{4} = \frac{3}{4}$. Taking the square root of both sides, we get.

26 [Foundational] • Target: Law of Sines and Law of Cosines

On the unit circle, a point $P(x, y)$ corresponds to an angle θ in standard position. If the angle $\theta = \frac{\pi}{3}$, what is the value of the y-coordinate of point P?

a. $\frac{1}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$ ✓

d. 1

The coordinates of a point on the unit circle corresponding to an angle θ are given by $(\cos\theta, \sin\theta)$. In this case, $\theta = \frac{\pi}{3}$. Therefore, the y-coordinate of point P is $\sin\left(\frac{\pi}{3}\right)$. We know that $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$.

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27 [Foundational] • Target: Trigonometric identities for compound angles

In triangle ABC, the lengths of sides a, b, and c are opposite to angles A, B, and C respectively. Given that $a = 7$, $b = 10$, and angle $C = 60^\circ$, find the exact length of side c.

a. $\sqrt{79}$

b. 9

c. $\sqrt{91}$ ✓

d. 10

We can use the Law of Cosines to find the length of side c. The Law of Cosines states that $c^2 = a^2 + b^2 - 2ab\cos(C)$. Substituting the given values: $c^2 = 7^2 + 10^2 - 2(7)(10)\cos(60^\circ)$ $c^2 = 49 + 100 - 140(\frac{1}{2})$ $c^2 = 149 - 70$ $c^2 = 79$ Therefore, $c = \sqrt{79}$.

28 [Foundational] • Target: A well-designed word problem: the height of a lighthouse

In a right-angled triangle ABC, where the right angle is at C, if $AC = 10$ and $BC = 24$, find the length of the hypotenuse AB.

a. 25

b. **26** ✓

c. 27

d. 28

We can use the Pythagorean theorem to find the length of the hypotenuse in a right-angled triangle. The theorem states that $a^2 + b^2 = c^2$, where a and b are the lengths of the two shorter sides (legs) and c is the length of the hypotenuse. In this triangle, $AC = 10$ and $BC = 24$. Therefore, we have: $AB^2 = AC^2 + BC^2$ $AB^2 = 10^2 + 24^2$ $AB^2 = 100 + 576$ $AB^2 = 676$ $AB = \sqrt{676}$ $AB = 26$ The length of the hypotenuse AB is 26.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Right-triangle trigonometry

In a right-angled triangle ABC, the angle B is 90 degrees. If the length of side AB is 8 units and the length of side BC is 6 units, what is the length of the hypotenuse AC?

a. 10 units ✓

b. 14 units

c. 48 units

d. 20 units

This is a right-angled triangle, so we can use the Pythagorean theorem: $a^2 + b^2 = c^2$. Here, the sides are $AB = 8$ and $BC = 6$, and the hypotenuse is AC. So, $AC^2 = AB^2 + BC^2$. $AC^2 = 8^2 + 6^2$ $AC^2 = 64 + 36$ $AC^2 = 100$ $AC = \sqrt{100}$ $AC = 10$ units. Common mistakes include adding the sides (8 + 6 = 14) or multiplying them (8 * 6 = 48) instead of applying the Pythagorean theorem.

30 [Intermediate] • Target: Exact values and the unit circle

A sector of a circle of radius $r = 8$ cm has central angle $\theta = 0.9$ rad. What is the area of the sector?

a. $28.8\ \text{cm}^2$ ✓

b. $57.6\ \text{cm}^2$

c. $33.8\ \text{cm}^2$

d. $7.2\ \text{cm}^2$

Sector area formula: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(8)^2(0.9) = 28.8\ \text{cm}^2$.

31 [Intermediate] • Target: Graphs of trigonometric functions

In ΔABC, $\angle A = 30^\circ$, $\angle B = 90^\circ$, and side $a = 10$. Find side b (Law of Sines).

a. 10

b. 22

c. 5

d. 20 ✓

$\frac{a}{\sin A} = \frac{b}{\sin B} \Rightarrow b = \frac{10 \sin 90^\circ}{\sin 30^\circ} = 20$.

32 [Intermediate] • Target: Trigonometric identities

A right triangle has legs 12 and 16 and hypotenuse 20. The angle θ is opposite the side of length 12. Find $\cos(\theta)$.

a. $\frac{4}{3}$

b. $\frac{3}{5}$

c. $\frac{3}{4}$

d. $\frac{4}{5}$ ✓

$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{16}{20} = \frac{4}{5}$.

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33 [Intermediate] • Target: Solving trigonometric equations

A right triangle has legs of length 18 and 24. Find the length of the hypotenuse.

a. 32

b. 34

c. 33

d. **30** ✓

By Pythagoras: $18^2 + 24^2 = 30^2$, so the unknown side is 30.

34 [Intermediate] • Target: Law of Sines and Law of Cosines

The function f is defined on the open interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ by $f(x) = \frac{2-\sin x}{\cos x} + \frac{\cos x}{2-\sin x}$. Determine the exact minimum value of $f(x)$.

a. $\frac{4\sqrt{3}}{3}$ ✓

b. 2

c. $\frac{5}{2}$

d. $\frac{3\sqrt{3}}{2}$

Let $u(x) = \frac{2-\sin x}{\cos x}$. For $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, we have $\cos x > 0$ and $2 - \sin x > 0$, so $u(x) > 0$. The given expression is of the form $g(u) = u + \frac{1}{u}$. **Step 1: Check feasibility of the standard AM-GM minimum** By the AM-GM inequality, $u + \frac{1}{u} \geq 2$, with equality if and only if $u = 1$. Setting $u = 1$: $\frac{2-\sin x}{\cos x} = 1 \Rightarrow \sin x + \cos x = 2$. Using the harmonic R-addition form.

35 [Intermediate] • Target: Trigonometric identities for compound angles

In triangle ABC, the side lengths are $AB = (x+2)$ cm, $BC = (3x-1)$ cm, and $AC = 7$ cm. Given that $\angle ABC = 60^\circ$ and $x > 0$, find the exact area of triangle ABC.

a. $10\sqrt{3}$ cm^2 ✓

b. $5\sqrt{3}$ cm^2

c. $20\sqrt{3}$ cm^2

d. 10 cm^2

Step 1: Use the cosine rule on triangle ABC to set up an equation for x: $AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos(\angle ABC)$ Substitute the given values and $\cos(60^\circ) = \frac{1}{2}$: $7^2 = (x+2)^2 + (3x-1)^2 - 2(x+2)(3x-1)(\frac{1}{2})$ Step 2: Expand each term: $(x+2)^2 = x^2 + 4x + 4$ $(3x-1)^2 = 9x^2 - 6x + 1$ $(x+2)(3x-1) = 3x^2 + 5x - 2$ Substitute these back into the equation: $49 = (x^2 + 4x + 4) + (9x^2 - 6x + 1) - (3x^2 + 5x - 2)$ $49 = 10x^2 - 2x + 5 - 3x^2 - 5x + 2$.

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36 [Intermediate] • Target: A well-designed word problem: the height of a lighthouse

Consider the complex number $z = \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{10}$. If z is expressed in polar form as $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $0 \le \theta < 2\pi$, find the exact values of r and θ.

a. $r = 1, \theta = \frac{5\pi}{3}$ ✓

b. $r = 1, \theta = \frac{4\pi}{3}$

c. $r = 10, \theta = \frac{5\pi}{3}$

d. $r = \sqrt{3}, \theta = \frac{4\pi}{3}$

First, express the base complex number in polar form. Let $w = \frac{\sqrt{3}}{2} + \frac{1}{2}i$. The modulus of w is $|w| = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{3}{4} + \frac{1}{4}} = \sqrt{1} = 1$. The argument of w is $\arg(w) = \arctan\left(\frac{1/2}{\sqrt{3}/2}\right) = \arctan\left(\frac{1}{\sqrt{3}}\right)$. Since the real and imaginary parts are both positive, w is in the first quadrant, so $\arg(w) = \frac{\pi}{6}$. Thus.

37 [Intermediate] • Target: Right-triangle trigonometry

In triangle ABC, $AB = 6$ cm, $AC = 10$ cm, and the area of triangle ABC is $15\sqrt{3}$ cm^2. Given that angle BAC is obtuse, find the exact length of BC.

a. 14 cm ✓

b. $2\sqrt{19}$ cm

c. $2\sqrt{34}$ cm

d. 8 cm

1. Use the formula for the area of a triangle: Area $= \frac{1}{2}(AB)(AC)\sin(\angle BAC)$ $15\sqrt{3} = \frac{1}{2}(6)(10)\sin(\angle BAC) = 30\sin(\angle BAC)$ $\sin(\angle BAC) = \frac{15\sqrt{3}}{30} = \frac{\sqrt{3}}{2}$ 2. Since $\angle BAC$ is obtuse ($90^\circ < \angle BAC < 180^\circ$): $\angle BAC = 180^\circ - 60^\circ = 120^\circ$ 3. Determine $\cos(\angle BAC)$: $\cos(120^\circ) = -\cos(60^\circ) = -\frac{1}{2}$ 4. Apply the cosine rule to find BC.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Exact values and the unit circle • Cross-Lesson: 4.01 Coordinate Geometry

A sector of a circle of radius 12 cm has a central angle of $\frac{\pi}{3}$ radians. Find the perimeter of the sector, giving your answer in exact form.

a. $(24 + 4\pi)$ cm ✓

b. $(12 + 4\pi)$ cm

c. 4π cm

d. $(24 + 24\pi)$ cm

To find the perimeter of a sector, we add the arc length to the two bounding radii. 1. Calculate the arc length s: $s = r\theta = 12 \times \frac{\pi}{3} = 4\pi$ cm 2. Calculate the perimeter P: $P = 2r + s = 2(12) + 4\pi = 24 + 4\pi$ cm Analysis of incorrect choices: - Choice B: Adds only one radius to the arc length, $r + s = 12 + 4\pi$. - Choice C: Computes only the arc length $s = 4\pi$, failing to include the two bounding radii. - Choice D: Incorrectly adds the full circumference.

39 [Challenging] • Target: Graphs of trigonometric functions • Cross-Lesson: 6.01 Differentiation & Integration

In the given right-angled triangle ABC, angle B is 90 degrees. Point D is on BC such that BD = 4 and DC = 9. If AD is the altitude to the hypotenuse AC, find the length of AD.

a. 6 ✓

b. 8

c. 12

d. 15

In a right-angled triangle, the altitude to the hypotenuse creates two smaller triangles that are similar to the original triangle and to each other. \\ According to the Geometric Mean Theorem, the square of the altitude to the hypotenuse is equal to the product of the lengths of the two segments it divides the hypotenuse into. \\ In triangle ABC, AD is the altitude to the hypotenuse BC. Therefore, we have: \\ $AD^2 = BD \times DC$ \\ Given that BD = 4 and DC = 9: \\ $AD^2 = 4 \times 9$ \\ $AD^2 = 36$ \\.

40 [Challenging] • Target: Trigonometric identities • Cross-Lesson: 7.01 Statistics & Probability

In the right-angled triangle ABC, shown below, AD is the altitude to the hypotenuse BC. If BD = 4 and DC = 9, find the length of the altitude AD.

a. 5

b. **6** ✓

c. 7

d. 8

In a right-angled triangle, the altitude to the hypotenuse creates similar triangles. By the Geometric Mean Theorem, the square of the altitude to the hypotenuse is equal to the product of the lengths of the two segments it divides the hypotenuse into. Thus, $AD^2 = BD \times DC$. Given $BD = 4$ and $DC = 9$, we have $AD^2 = 4 \times 9 = 36$. Taking the square root of both sides, we get $AD = \sqrt{36} = 6$.

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41 [Challenging] • Target: Solving trigonometric equations • Cross-Lesson: 8.01 Functions: Composite & Inverse

In ΔABC, $a = 8$, $b = 10$, $c = 12$. Find $\angle A$ (to one decimal place) using the Law of Cosines.

a. **41.4°** ✓

b. 46.4°

c. 138.6°

d. 36.4°

By the Law of Cosines: $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{100 + 144 - 64}{240} \approx 0.75$, so $A = \cos^{-1}(0.75) \approx 41.4°$.

42 [Challenging] • Target: Law of Sines and Law of Cosines • Cross-Lesson: 9.01 Circular Measure

In ΔABC, $a = 5$, $b = 6$, $c = 7$. Find $\angle A$ (to one decimal place) using the Law of Cosines. (trial 2)

a. 49.4°

b. 135.6°

c. **44.4°** ✓

d. 39.4°

By the Law of Cosines: $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{36 + 49 - 25}{84} \approx 0.714$, so $A = \cos^{-1}(0.714) \approx 44.4°$.

43 [Challenging] • Target: Trigonometric identities for compound angles • Cross-Lesson: 1.01 Indices, Surds & Quadratics

If $\cos\theta = \frac{1}{2}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{1}{4}$

b. $\frac{1}{2}$ ✓

c. $\frac{3}{5}$

d. $-\frac{1}{2}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1 - \cos\theta}{2}} = \sqrt{\frac{1 - \left(\frac{1}{2}\right)}{2}} = \frac{1}{2}$.

44 [Challenging] • Target: A well-designed word problem: the height of a lighthouse • Cross-Lesson: 10.01 Series & Binomial Theorem

In ΔABC, $b = 10$, $c = 7$, and $\angle A = 120°$. Find side a (to one decimal place) using the Law of Cosines.

a. 13.8

b. **14.8** ✓

c. 15.8

d. 12.2

By the Law of Cosines: $a^2 = b^2 + c^2 - 2bc\cos A = 10^2 + 7^2 - 2(10)(7)\cos 120° \approx 219.0$, so $a \approx 14.8$.

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45 [Challenging] • Target: Right-triangle trigonometry • Cross-Lesson: 11.01 Kinematics

In triangle ABC, $a = 5$, $b = 10$, and $C = 120^\circ$. Find the exact length of c.

a. $5\sqrt{7}$ ✓

b. $\sqrt{179}$

c. 15

d. $3\sqrt{19}$

$c^2 = a^2 + b^2 - 2ab\cos C = (5)^2 + (10)^2 - 2(5)(10)\cos(120^\circ) = 175 \Rightarrow c = 5\sqrt{7}$.

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